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Cross-Domain SAR to RGB Image Translation using Conditional GAN with Perceptual Guidance

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***Abstract*— Synthetic Aperture Radar (SAR) images are an important source of remote sensing data because it can obtain high-resolution surface information regardless of weather and lighting conditions. SAR images are, however, grayscale and do not possess the same semantic richness as optical imagery so they are not easily understandable for human analysis. This paper proposes a method for SAR image colorization based on deep learning using a conditional Generative Adversarial Network (cGAN). The proposed architecture which utilizes a U-Net-based generator to map single-channel SAR images encourages local realism and texture consistency to check visual consistency. We use a composite loss function that combines perceptual loss calculated using pre-trained VGG19 features, pixel-wise L1 loss, and adversarial loss to enhance visual quality and structural accuracy. A dataset of SAR-optical image pairings spanning different land cover classes is used to train the model. Experimental outcomes show that the colorized images generated maintain fine spatial patterns and attain high visual realism, thereby making the method promising for real-world applications in Earth observation and remote sensing analysis.**

Keywords—Generative Adversarial Network (GAN), Synthetic Aperture Radar (SAR), Conditional GAN, CycleGAN, Image Colorization.

I. INTRODUCTION

Synthetic Aperture Radar (SAR) images are widely used in remote sensing and Earth observation. This is mainly due to its capacity to collect high-quality data regardless of light levels, cloud cover, or weather. Designed to function in the microwave band, SAR systems have the ability to image the Earth's surface during both day and night-time. These properties are the reason why SAR images have such wide applications such as environmental observation, flood mapping, land cover classification, and defense surveillance. In addition to these benefits, one of the inherent limitations of SAR imagery is that it does not have natural color. To overcome this limitation, there is growing interest in formulating sophisticated methods for SAR image colorization and cross-modal translation, which can fill the semantic gap between SAR and optical spaces.

Among these, deep learning-based methods, especially those that use Generative Adversarial Networks (GANs), have yielded encouraging results. Deep learning has become a strong paradigm for image colorization in the last few years. Amongst all the different methods, Generative Adversarial Networks (GANs) have proven extremely successful in creating high-fidelity optical-like images from SAR inputs [1, 5, 6, 7, 8, 9]. Pix2Pix, CycleGAN, and recently attention-guided and multi-scale GANs have facilitated unsupervised or semi-supervised learning to translate SAR images to optical image.

In this paper, we propose that combining perceptual loss with adversarial and pixel-wise losses can greatly enhance the visual realism and semantic accuracy of SAR image to optical or RGB image translation. We introduce a deep learning-based approach based on a conditional Generative Adversarial Network (cGAN), where a generator based on U-Net produces RGB images from grayscale SAR inputs and a PatchGAN discriminator helps ensure realism by discriminating between real and fake image pairs. In addition, we use a perceptual loss calculated from intermediate VGG19 features to maintain high-level texture and structure. We incorporate a composite loss function that includes various types of losses such as adversarial loss, L1 reconstruction loss, and perceptual loss, to enhance both pixel-level and perceptual quality.

II. RELATED WORK

In this section we examine several approaches developed for transforming SAR images into natural-looking optical images.

Seg-CycleGAN [1] utilizes semantic segmentation—evaluated on HRSID, DIOR, and WHU-OPT-SAR datasets—to guide the SAR image to optical image translation, yielding visually better and semantically consistent results. The approach expands the CycleGAN model by introducing a segmentation loss obtained from a pre-trained semantic segmentation module (learned from annotations of the Segment Anything Model (SAM) [2]). Seg-CycleGAN shows remarkable improvement over unpaired alternatives such as U-GAT-IT [3] and CUT [4]. MS-GAN [5] employs a multiscale generator (MRG) and a denoising module (SSD) (evaluated on SEN1-2 and WHU-SEN-City) to learn complex scenes. It effectively preserves scene details and reduces artifacts. S-CycleGAN [6] integrates the strengths of pix2pix

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and CycleGAN, using cycle-consistency loss and pixel-wise Mean Squared Error loss to preserve both structural and land cover information during translation. This process involves dividing big SAR images into patches, translating them with S-CycleGAN, and stitching the results to create a coherent optical image. Sar2color [7] is an image transformation model that converts a SAR image to optical image using a conditional GAN. It incorporates new modules such as DCT Residual Block and Light-ASPP to overcome speckle noise and geometry distortions. MC-GAN [8] (Multidomain Cycle-Consistency Generative Adversarial Network) is a new approach that takes advantage of a mask vector which is a terrain-specific one-hot encoding for colorization along with cycle-consistency loss to support bidirectional SAR-optical image mapping. The cGAN4ColSAR [9] model extends the Pix2Pix architecture with a U-Net based generator and PatchGAN discriminator. The U-Net architecture, is added with skip connections. The discriminator uses a patch-based strategy to improve high-frequency detail recovery. The proposed SAR-GAN [10], a cascaded architecture that consists of two sub-networks: despeckling CNN and colorization CNN. Both were trained together in an end-to-end manner. The colorization network uses an encoder-decoder architecture which has skip connections to preserve spatial details. Adversarial loss is incorporated to promote realistic conversion, while L1 loss guarantees pixel-level precision. Such an integrated approach allows for effective denoising and colorization of SAR images in a single process. The work [11] introduces a bidirectional SAR image to optical image translation model based on a Cascaded-Residual Adversarial Network (CRAN). The technique adopts a U-Net-based encoder-decoder generator along with multi-scale cascaded residual connections for enhanced texture and structure preservation. It adopts a hybrid GAN + L1 loss to enable the fusion of realism and structural accuracy. The model allows for both supervised and unsupervised training (through CycleGAN loops). The paper [12] used a SAR image to optical image translation model based on a color-supervised conditional diffusion approach. In contrast to GAN-based methods, it relies on DDPMs (Denoising Diffusion Probabilistic Model) for stable and high-fidelity generation. The SAR input is conditioned and the denoising process is directed at each step by a 1×1 convolution. A color supervision loss that uses Gaussian-blurred reconstructions is introduced to enhance visual realism, thus reducing color shift while maintaining structural consistency.

III. Methodology

This section details the proposed deep learning framework for colorizing grayscale SAR images into natural looking RGB images by using a conditional Generative Adversarial Network (cGAN). The overall architecture is made up of a U-Net-based generator, a PatchGAN discriminator, and a multi-component loss function that includes adversarial, pixel-wise, and perceptual losses as shown in Fig.1.

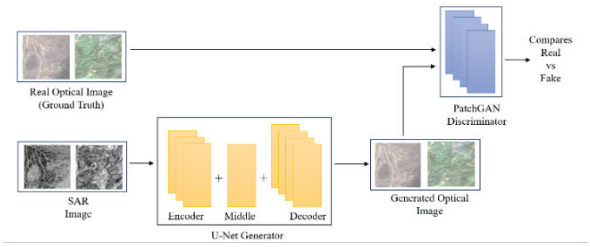


Fig.1. Flow diagram of proposed SAR-to-RGB translation model

This work seeks to translate SAR representations into the optical domain by reconstructing visually coherent images that preserve both appearance and spatial integrity. The architecture is tailored for multi-class SAR datasets with domain-specific textures such as agriculture, urban, barrenland, and grassland. We have used a combined loss function consisting of three components: adversarial (GAN) loss, L1 loss, and perceptual loss.

A. Dataset Preprocessing

The dataset used for this research is the SEN 1-2 dataset which consists of SAR images and their corresponding RGB images. These are labeled into four land cover classes: agriculture, urban, grassland, and barrenland. Each class directory has two subdirectories: one for SAR images (s1/) and one for corresponding RGB ground truth images (s2/). We partitioned the dataset into training and testing sets to enable both supervised learning and quantitative assessment.

All the input SAR images were resized to 256×256 pixels. The grayscale SAR images were normalized to a one-channel, and the RGB images were normalized to the range [−1,1] using Tanh activation function.

B. Generator

Our generator is a U-Net style. It has a fully convolutional encoder-decoder structure with skip connections. The architecture as presented in Fig. 2 allows efficient spatial feature propagation while learning translation from single-channel SAR (1×256×256) to RGB (3×256×256). The encoder extracts hierarchical features from the SAR input using a layered structure of convolutional blocks with Leaky ReLU activation and batch normalization. The decoder upsamples these features using transposed convolutions and integrates encoder features through concatenation.

It is made up of a down-sampling encoder, a bottleneck, and an upsampling decoder. The encoder operates on the input single-channel SAR grayscale image and continuously extracts hierarchical features through convolutional layers with a stride value of 2 for spatial down-sampling. Each convolutional layer has a Leaky ReLU activation function. Batch normalization is used from the second layer to enhance training stability and convergence. At the bottleneck, the network compresses the features into a concise latent representation via a 512-channel convolutional layer.

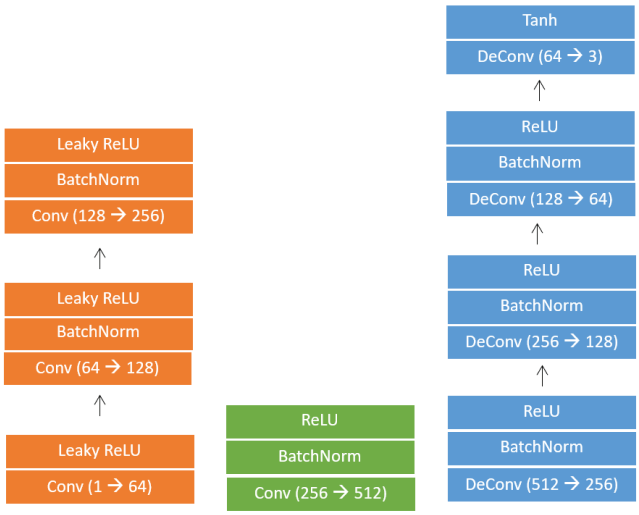


Fig.2. Architecture of U-Net Generator with Encoder, Bottleneck and Decoder

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The decoder then regenerates the colorized RGB images by iteratively upsampling the feature maps via transposed convolutional layers (or deconvolutions). A transposed convolution, ReLU activation, and batch normalization constitute the decoder block. The last decoder layer employs a transposed convolution to convert features to a 3-channel RGB image and applies a Tanh activation to convert the output values within the range [-1, 1]. U-Net architecture also helps improves reconstruction quality through the use of skip connections across encoder and decoder layers of the same resolution.

C. Discriminator

The discriminator follows the PatchGAN architecture, which evaluates the realism of local patches rather than the image as a whole. This architecture allows high-frequency detail in the output. The SAR image and its corresponding RGB image are concatenated and fed into the discriminator, thus producing a 4-channel input tensor. It passes this input through a layered structure of convolutional blocks as shown in Fig.3 with growing depth, each of which is followed by batch normalization and Leaky ReLU activation. The last layer produces a single-channel feature map corresponding to the probability that each patch in the image is real.

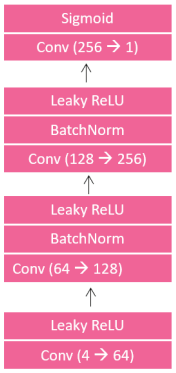


Fig.3. Architecture of Patch-GAN based D

D. Loss Functions

In order to train the proposed SAR-to-optical transfer network well, we use a combined loss function made up of three components: adversarial (GAN) loss, L1 loss, and perceptual loss. All three loss terms play different roles in driving the model towards high-quality, visually pleasing colorized results.

1) Adversarial Loss: The adversarial loss (L_{GAN}) is derived from the cGAN architecture. It trains the generator to generate outputs that are indiscernible from actual images for the discriminator, thus promoting photorealism.

2) Pixel-wise L1 Loss: We use an L1 loss (L_{L1}) that aims to reduce the mean absolute error (MAE) between predicted and real pixels to make sure the generated image structurally resembles the original RGB image.

3) Perceptual Loss: The perceptual loss (L_{perc}) encourages the generator to preserve high-level semantic and texture information by comparing feature representations from a pretrained VGG19 network.

4) Total Generator Loss: The overall loss for the generator is a weighted sum of the three generator-associated losses. (1) (here $\lambda_{L1} = 100$ and $\lambda_{perc} = 0.1$ are empirically chosen weights.)

$$L_G = L_{GAN} + \lambda_{L1} \cdot L_{L1} + \lambda_{perc} \cdot L_{perc} \quad (1)$$

5) Discriminator Loss: The discriminator loss keeps the discriminator in check so that it predicts real pairs as real and fake pairs as fake correctly. This is a binary cross-entropy loss over real vs. fake predictions.

IV. Results and Discussion

This section describes the qualitative and quantitative comparison of the proposed SAR image to RGB image translation model. The performance is compared on several land cover classes (Urban, Grassland, Barrenland, and Agricultural) to analyze the generalizability and visual quality of generated images. Visual comparisons and numerical measurements are used to highlight the merits of the method. In order to test the realism and fidelity of the produced RGB images, we employ a combination of losses observed during training (L1, Adversarial (GAN), Perceptual) and visual inspection over land cover. The outcomes are examined in detail preservation, perceptual correctness, and training stability.

A. Training Loss Curves

The curves of training loss help understand how well the model has learned over epochs. We trained the model for 100 epochs. The training losses include L1 Loss (Fig.4), Adversarial (GAN) Loss (Fig. 5), Perceptual Loss (Fig. 6), Generator Loss (Fig. 7), and Discriminator Loss (Fig. 8).

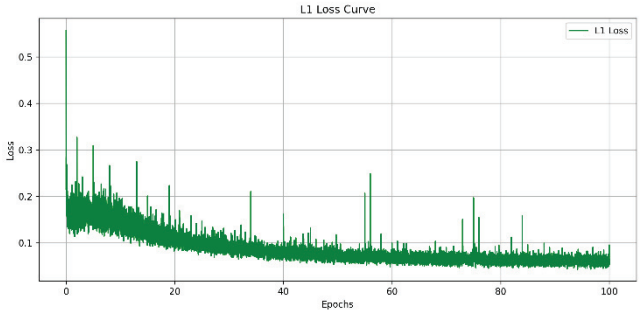


Fig.4. Epoch-wise L1 loss

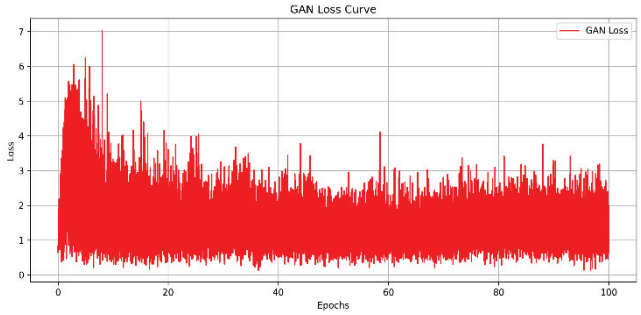


Fig.5. Epoch-wise Adversarial (GAN) Loss Curve

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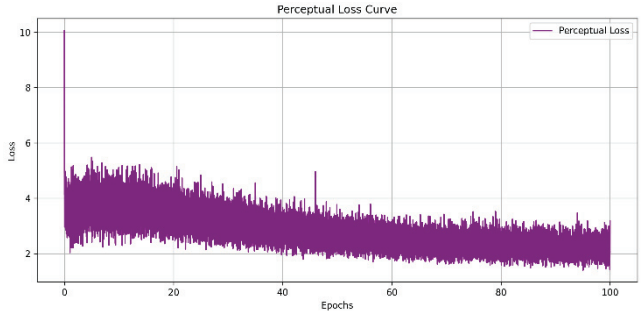


Fig.6. Epoch-wise Perceptual Loss Curve

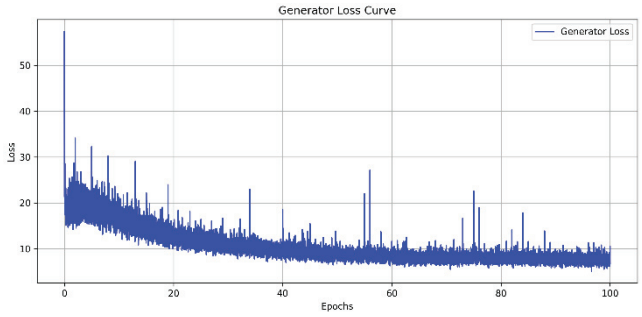


Fig.7. Epoch-wise Generator Loss Curve

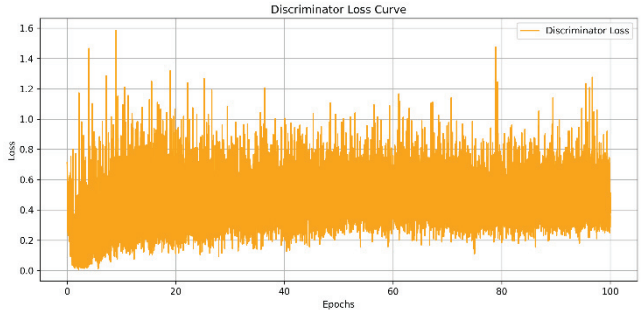


Fig.8. Epoch-wise Discriminator Loss Curve

All the training loss curves are displayed together in one plot in Fig. 9. From the loss curves, it is evident that all components of the model converged progressively over the training epochs.

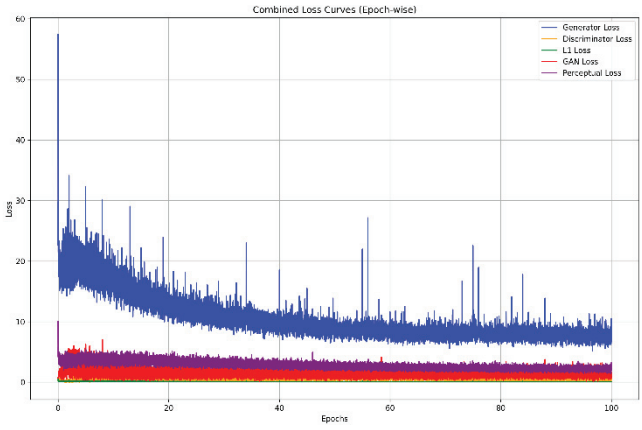


Fig.9. Epoch-wise Combined Loss Curve

Both L1 loss and perceptual loss indicate a stable decrease over time, which shows better reconstruction at the pixel level and feature level. The generator loss declined gradually, which emphasizes the improved capability of the model to generate realistic images. While the GAN loss showed some variation, it stayed within a healthy range, which is an expected result of adversarial training. Similarly, the discriminator loss remained low and consistent, indicating that the discriminator was performing well in distinguishing real and generated pairs.

B. Qualitative Evaluation by Land Cover Class

To assess the visual quality of the output our model, we produced RGB outputs from single-channel SAR inputs for four land cover classes: Urban, Grassland, Barrenland, and Agricultural shown in Fig.10-21 (Image on the left side is output image and image on the right side is Ground Truth). Across all classes, the model demonstrates strong semantic understanding and gives visibly coherent results.

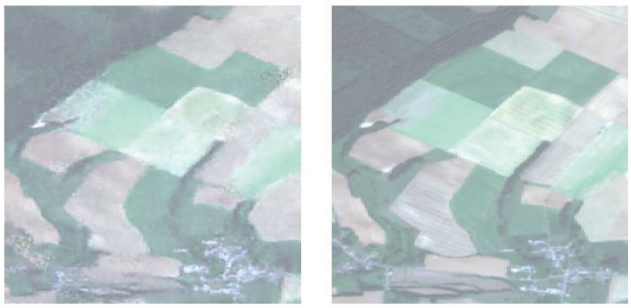


Fig.10. Agricultue Class Result Sample 1

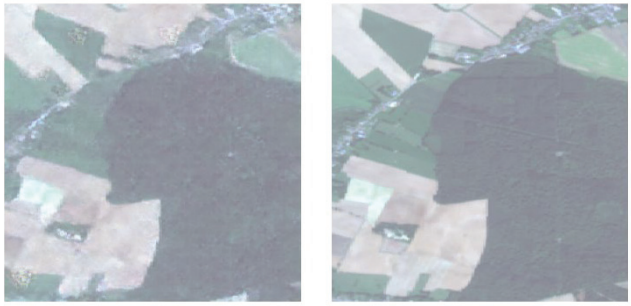


Fig.11. Agricultue Class Result Sample 2

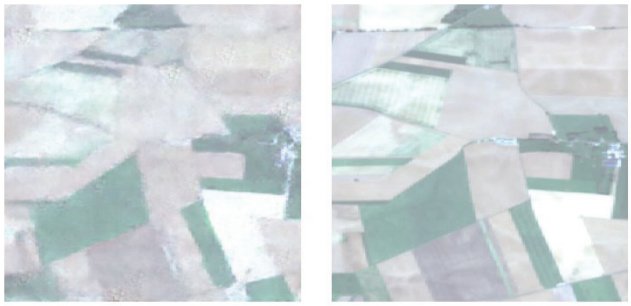


Fig.12. Agricultue Class Result Sample 3

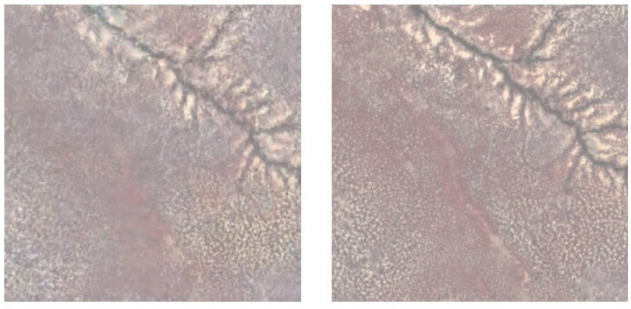


Fig.13. Barrenland Class Result Sample 1

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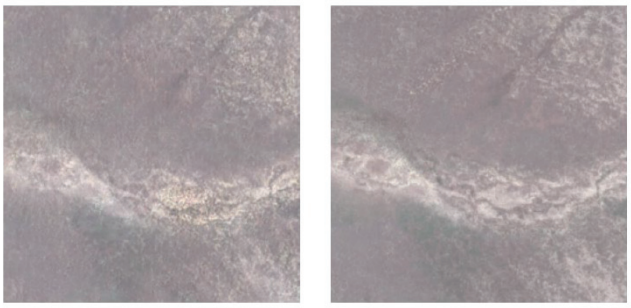


Fig.14. Barrenland Class Result Sample 2

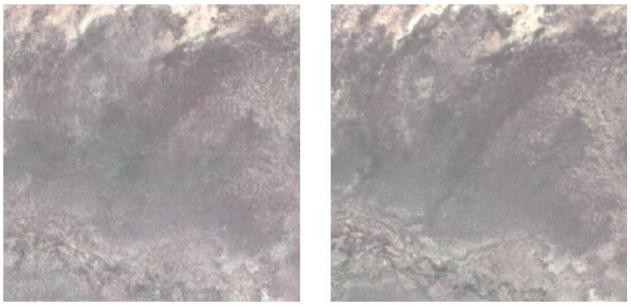


Fig.15. Barrenland Class Result Sample 3

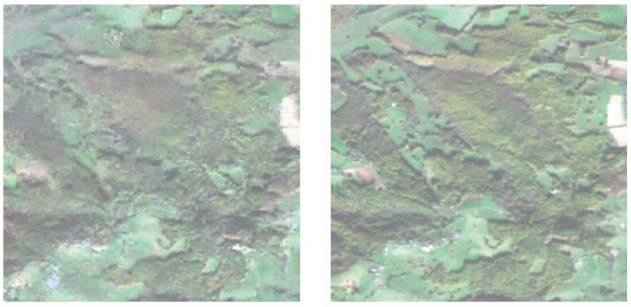


Fig.16. Grassland Class Result Sample 1

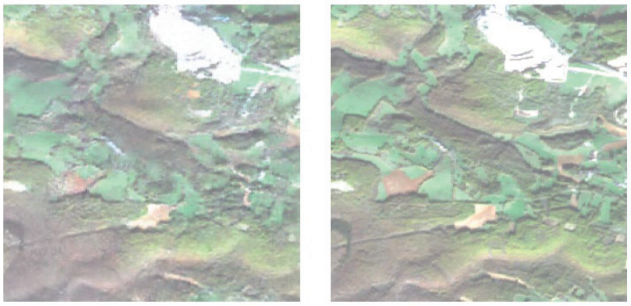


Fig.17. Grassland Class Result Sample 2

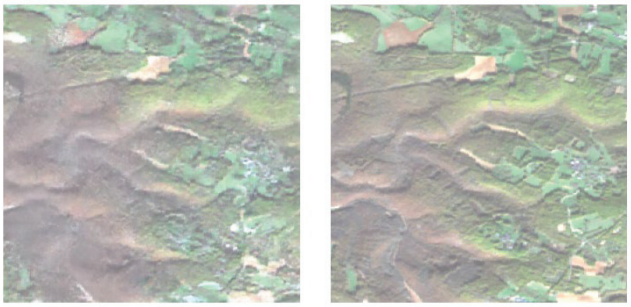


Fig.18. Grassland Class Result Sample 3



Fig.19. Urban Class Result Sample 1



Fig.20. Urban Class Result Sample 2



Fig.21. Urban Class Result Sample 3

C. *Quantitative Evaluation*

To better check how well the model is working, we used two common measures for image similarity: PSNR and SSIM. The proposed model recorded an SSIM score of 0.71 on average, implying that the model effectively retains spatial and structural features of the target images. A noted average PSNR of 27.46 dB indicated decent image quality and negligible noise in generated outputs. The comparative study of different SAR image colorization models is presented in Table I. We further calculated the PSNR and SSIM of each class (Urban, Grassland, Barrenland, and Agricultural) to understand which class has the best results as presented in Table II.

TABLE I. COMPARITIVE STUDY OF SAR IMAGE COLORIZATION MODELS BASED ON PSNR AND SSIM

Ref. No.	PSNR and SSIM Scores of Different models		
	Model Name	***PSNR***	***SSIM***
1	Seg-Cycle-GAN	19.259	0.346
5	MS-GAN	14.62	0.308
6	S-Cycle-GAN	11.203	0.377
7	Sar2Color	18.8145	0.4162
8	MC-GAN	19.132	0.416
	Our Model	**27.46**	**0.71**

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TABLE II. CLASS-WISE EVALUATION OF IMAGES GENEARATED BBY OUR MODEL USING PSNR AND SSIM

Sr. No.	PSNR and SSIM Scores by Land Cover Class		
	Class Name	***PSNR***	***SSIM***
1	Agriculture	29.66	0.78
2	Barrenland	27.92	0.66
3	Grassland	25.28	0.64
4	Urban	26.98	0.76

The model showed the best results with the Agriculture class with the highest PSNR value of 29.66 dB and an SSIM value of 0.78, which implies high fidelity and perceptual similarity to the ground truth. The Urban class closely follows with PSNR of 26.98 dB and SSIM of 0.76, which is most likely due to the dominant geometric structures and regular patterns within cityscapes. Conversely, the Barrenland and Grassland classes have comparatively lower scores, with PSNR values of 27.92 dB and 25.28 dB, and SSIM scores of 0.66 and 0.64, respectively. These values are lesser and could be due to the absence of clear textures and uniform surface textures, which complicates colorization more in these types.

V. CONCLUSION

In this paper, we introduced a deep learning-based SAR image colorization approach based on a U-Net-based generator and patch-GAN-based discriminator, trained with adversarial, pixel-wise (L1), and perceptual losses. The model successfully translates single-channel SAR images into perceptually realistic RGB images, as indicated by both qualitative visual results and quantitative measures. The class-based assessment reveals that the model is strongest on organized areas such as agricultural and urban classes, and performance is relatively weaker on texture-deficient areas such as barrenland and grassland. As a whole, the findings support the possibility of using GAN-based architectures for cross-modal translation in remote sensing, and the new approach provides a promising direction towards rendering SAR imagery interpretable for human investigation and downstream vision tasks.

REFERENCES

[1] H. Zhang et al., "Seg-CycleGAN: SAR-to-Optical Image Translation Guided by a Downstream Task," in IEEE Geoscience and Remote Sensing Letters, vol. 22, pp. 1-5, 2025, Art no. 4005205, doi: 10.1109/LGRS.2025.3538868.

[2] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollar, and R. Girshick, “Segment anything,” in Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), October 2023, pp. 4015–4026.

[3] J. Kim, M. Kim, H. Kang, and K. Lee, “U-gat-it: Unsupervised generative attentional networks with adaptive layer-instance normalization for image-to-image translation,” arXiv preprint arXiv:1907.10830, 2019.

[4] T. Park, A. A. Efros, R. Zhang, and J.-Y. Zhu, “Contrastive learning for unpaired image-to-image translation,” in Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part IX 16. Springer, 2020, pp. 319–345.

[5] Z. Guo, Z. Zhang, Q. Cai, J. Liu, Y. Fan and S. Mei, "MS-GAN: Learn to Memorize Scene for Unpaired SAR-to-Optical Image Translation," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 17, pp. 11467-11484, 2024, doi: 10.1109/JSTARS.2024.3411691.

[6] L. Wang et al., "SAR-to-Optical Image Translation Using Supervised Cycle-Consistent Adversarial Networks," in IEEE Access, vol. 7, pp. 129136-129149, 2019, doi: 10.1109/ACCESS.2019.2939649.

[7] Guo, Zhe, Haojie Guo, Xuewen Liu, Weijie Zhou, Yi Wang, and Yangyu Fan. 2022. "Sar2color: Learning Imaging Characteristics of SAR Images for SAR-to-Optical Transformation" Remote Sensing 14, no. 15: 3740. https://doi.org/10.3390/rs14153740.

[8] G. Ji, Z. Wang, L. Zhou, Y. Xia, S. Zhong and S. Gong, "SAR Image Colorization Using Multidomain Cycle-Consistency Generative Adversarial Network," in IEEE Geoscience and Remote Sensing Letters, vol. 18, no. 2, pp. 296-300, Feb. 2021, doi: 10.1109/LGRS.2020.2969891.

[9] [9] K. Shen, G. Vivone, X. Yang, S. Lolli, and M. Schmitt, "A benchmarking protocol for SAR colorization: From regression to deep learning approaches," Neural Networks, vol. 169, pp. 698–712, 2024. [Online]. Available: https://doi.org/10.1016/j.neunet.2023.10.058.

[10] P. Wang and V. M. Patel, “Generating High Quality Visible Images from SAR Images Using CNNs,” *arXiv preprint arXiv:1802.10036*, 2018. [Online]. Available: https://doi.org/10.48550/arXiv.1802.10036.

[11] S. Fu, F. Xu, and Y.-Q. Jin, “Reciprocal Translation between SAR and Optical Remote Sensing Images with Cascaded-Residual Adversarial Networks,” *arXiv preprint arXiv:1901.08236*, 2019. [Online]. Available: https://doi.org/10.48550/arXiv.1901.08236.

[12] X. Bai and F. Xu, “SAR to Optical Image Translation with Color Supervised Diffusion Model,” *arXiv preprint arXiv:2407.16921*, 2024. [Online]. Available: https://doi.org/10.48550/arXiv.2407.1692.

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